

Information
Visualization

Preliminary concepts

Lesson 2

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01

Research and design issues

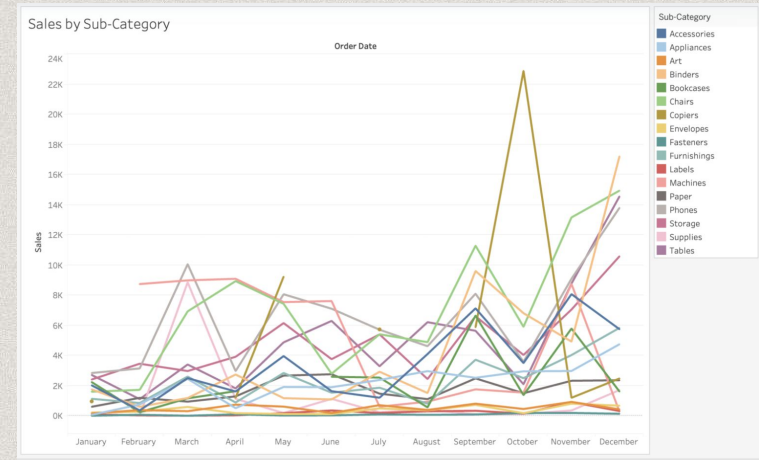
Terminology, expectations and best practices do
design effective visualisations

Types of analysis

Effective visualisations are able to identify, **summarise** and prioritise information.

The visual summary should be allow one to perform (either or both):

- **exploratory data analysis (EDA)**
Overview of an attribute
- **Knowledge Discovery (KD)**
Hypothesis-driven answer to a question



An example of graph that identifies and summarises data, but struggles to prioritise information.

Exploratory Data Analysis

It assumes that what is inside a dataset is not known, and one must still figure out what its added value is.

What is the dataset good for? which types of analyses does it enable?

In results presentations, EDA is used as introduction and summary, to introduce readers to the data and the phenomenon addressed.



Knowledge discovery

It's hypothesis-driven analysis.

The designer has a question to answer, a hypothesis, and wants to demonstrate it.

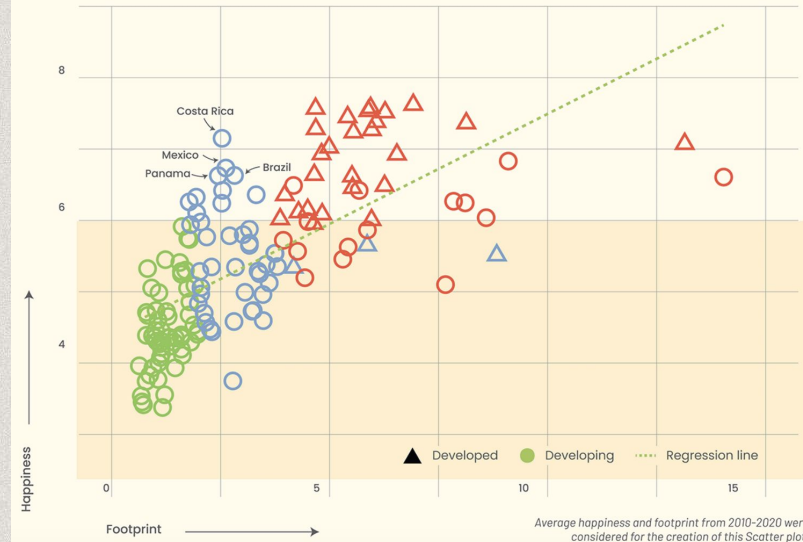
The validation consists of the **visual evidence** of a trend, a behaviour, an outlier, or a relationship (i.e. a pattern).

In results presentations, it's the climax of the narrative.

Sustainable Happiness

WHAT WE CAN LEARN FROM THE COUNTRIES THAT DON'T FOLLOW THE TREND: DEVELOPED VS. DEVELOPING

Countries like **Costa Rica, Mexico, Panama, and Brazil** all have low ecological footprints but are just as happy as the more developed European countries. What is it that Central America is doing that helps to find the balance between sustainability and happiness?



source

Results of data analysis

Results of a data-driven analysis (i.e. the interpretation of visuals)
MUST be:

- **Valid:** hold true when new data are added (representative if completeness cannot be ensured).
- **Novel*:** non-obvious.
- **Actionable:** usable in practical tasks (e.g. decision-making).
- **Understandable:** interpretable and meaningful to humans.

Expectations on novelty

Outcomes of data analysis/discovery techniques fall into two dichotomies: **expected/unexpected, positive/negative**

Health and a good diet are positively correlated

Health and junk food are negatively correlated

Expected positive	Unexpected negative
Expected negative	Unexpected positive

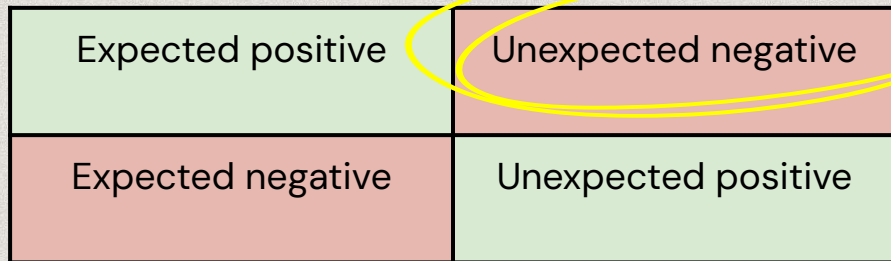
Gender affects the chances to get a loan

Playing with less toys leads children to play longer

Expectations on novelty

Unexpected negative results are the most insightful, **actionable** patterns to support decision-making (something you can act upon to change the future).

Expected positive	Unexpected negative
Expected negative	Unexpected positive

A 2x2 matrix diagram with a yellow circle highlighting the 'Unexpected negative' cell. The matrix is divided into four quadrants by a vertical and a horizontal line. The top-left quadrant is light green and contains the text 'Expected positive'. The top-right quadrant is light red and contains the text 'Unexpected negative'. The bottom-left quadrant is light red and contains the text 'Expected negative'. The bottom-right quadrant is light green and contains the text 'Unexpected positive'. A yellow circle is drawn around the top-right quadrant, which is the 'Unexpected negative' cell.

Negative results

A terminological mismatch.

Negative (or null) results refer to those insights that contradict the initial hypothesis or that do not show enough statistical significance to confirm/deny a hypothesis.

Possible reasons:

- The original hypothesis was incorrect
- The findings cannot be replicated
- Technical problems

While they have often been discredited by the scientific community (it's difficult to publish them, especially if data don't support an interesting alternative hypothesis) they provide interesting insights that could influence future analyses.

False/True

Positive/Negative results

	Positive	Negative
False	The analysis demonstrates that a condition holds while it actually does not.	The analysis demonstrates that a condition does not to hold while it actually does.
True	The analysis demonstrates that a condition holds and it actually does.	The analysis demonstrates that a condition does not hold and it actually does not.

Examples

Negative result.

A hypothesis we thought was true is demonstrated being wrong.

False negative.

A hypothesis demonstrated being wrong is actually true.

False positive.

A hypothesis demonstrated being true is actually wrong.

1. A research shows that during COVID, whether you were an avid video game player or not, this factor did not affect your mental health overall.
2. Another research, based on a **different sample**, shows that video games did affect your mental health. Study n.1 was a false negative.
3. Another research, that has a **stronger methodological approach** in selecting the study population, shows that video games do not affect you. Study n.2 is a false positive.

Design issues

In the previous class we saw the Tidwell's variables, i.e. how perception and interpretation are affected by visual features.

The interpretation of visual aids is not objective, it depends on the **reader's background**. But it can be influenced also by the **designer's design choices**.

E.g. the Halo effect

People use one positive attribute to infer other qualities. Researchers have demonstrated a causality link between beautiful visualisations and trust in their content [1,2].

Even the thickness of a bar can affect perception.

Edward Tufte's principles

There are several aspects that designer can tweak to affect the perception of visuals.

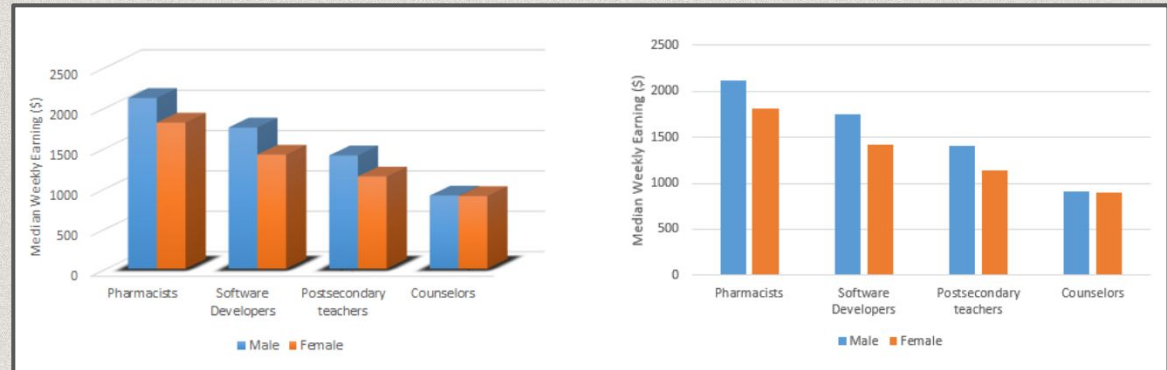
Edward Tufte designed a set of good practices to be taken into account when **finalising a visualization**.

Skiena S. Data Science Design Manual. Springer. 2017

Edward Tufte's principles

Minimize ink-ratio

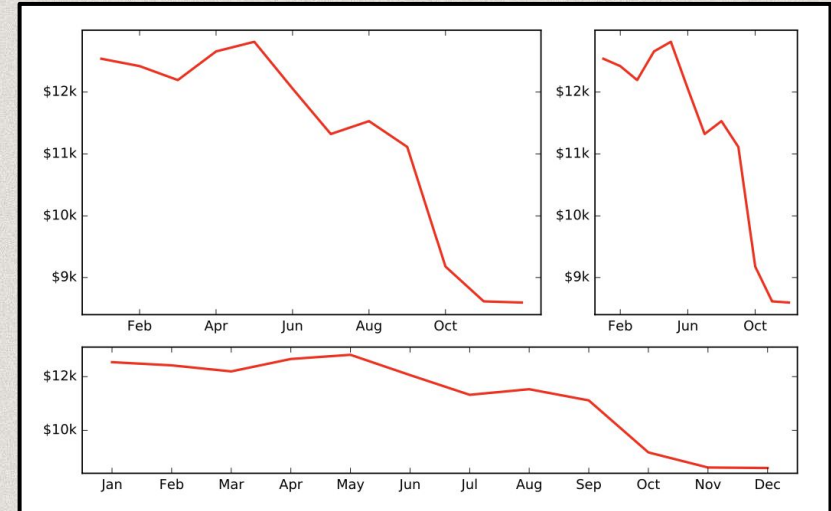
The ink used to represent data should be minimal, so that the intended message is clear.



Edward Tufte's principles

Minimize lie-factor

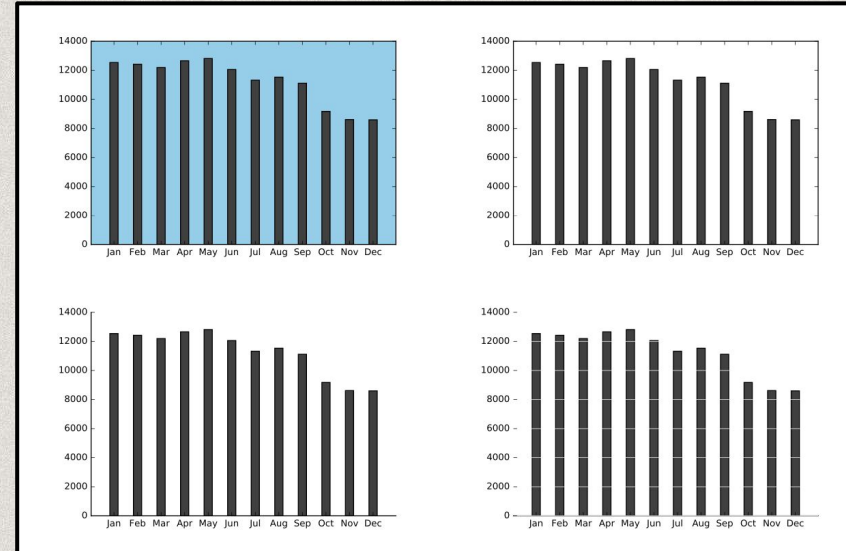
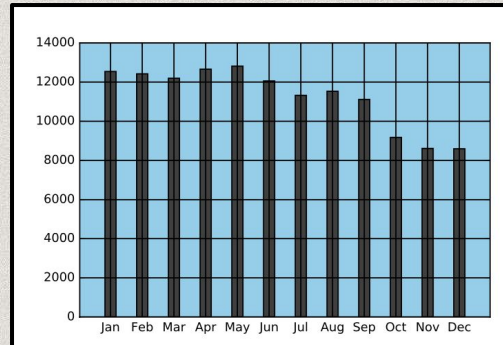
Omitting data or
changing proportions in
the visualisation can
tell different stories.



Edward Tufte's principles

Minimize chartjunk

Visual
elements
should not
hide the
data.



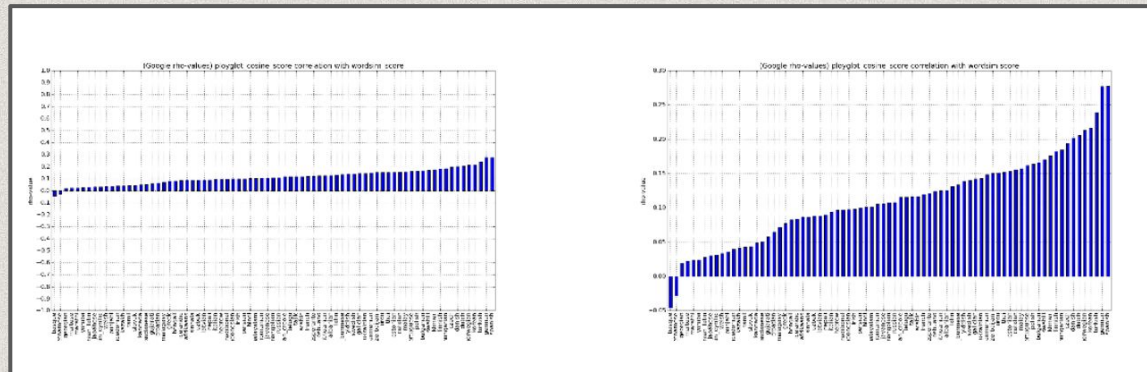
Edward Tufte's principles

Scale and ratio

Data should be scaled so as to fill all the space allotted to it.

When comparing charts using different scales you may mislead interpretation.

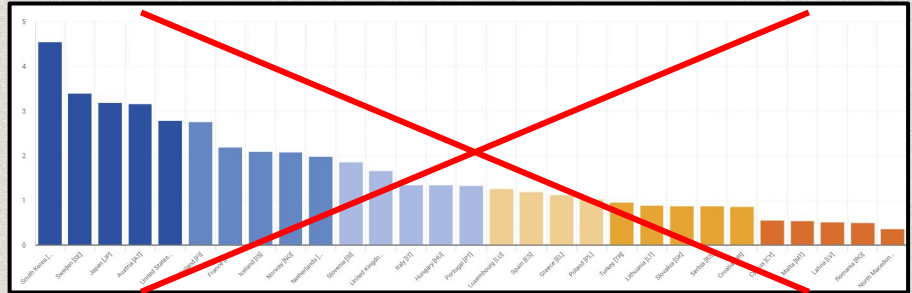
Labels should be in high resolution.



Edward Tufte's principles

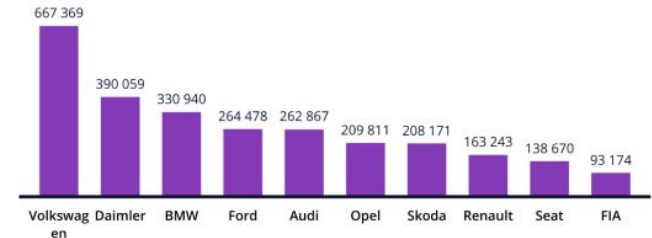
Use of color

Use colors to distinguish values and represent **difference** - rather than representing linear, numerical scales.



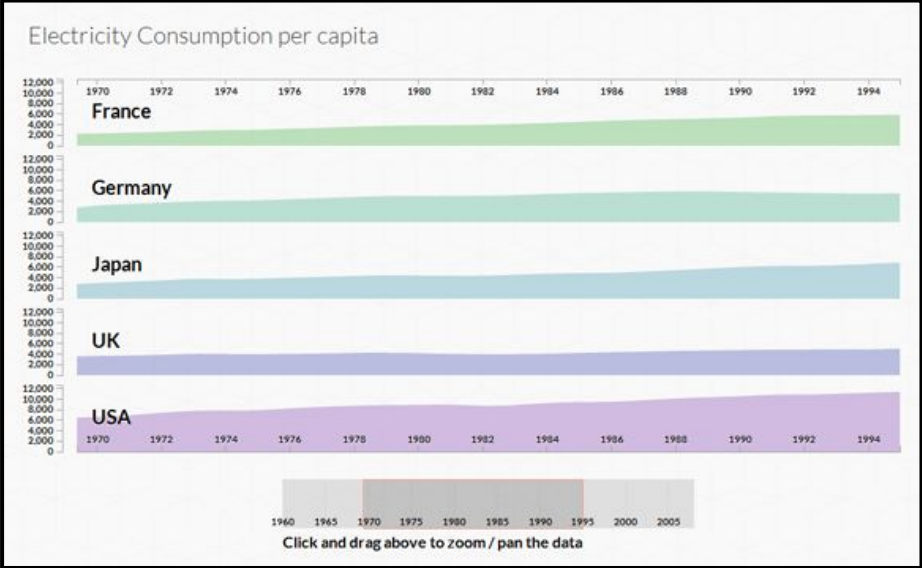
Car sales in Germany

Top10 sellers in 2019



Edward Tufts

Use multiple smaller charts
to facilitate comparison
(while losing the big
picture)





02

Charts and graphs

An inspirational journey rather than a complete survey

Data Visualization types

In the literature you will find many categorizations of charts (according to data types, number of variables, level of interactivity, task, affordance, etc.)

Tables

Graphs

Charts

Maps

Networks

Some catalogues

- Atlas
- Dataviz

Tables

Demonstrate results

If you want to scientifically **demonstrate** something, you will be asked to show a table.

It's often the **starting point** to realize more sophisticated visualizations.

0.103	0.176	0.387	0.300	0.379	0.276	0.179	0.321	0.192	0.250
0.333	0.384	0.564	0.587	0.857	1.064	0.698	0.621	0.232	0.316
0.421	0.309	0.654	0.729	0.228	0.529	0.832	0.935	0.452	0.426
0.266	0.750	1.056	0.936	0.911	0.820	0.723	1.201	0.935	0.819
0.225	0.326	0.643	0.337	0.721	0.837	0.682	0.987	0.984	0.849
0.187	0.586	0.529	0.340	0.829	0.835	0.873	0.945	1.103	0.710
0.153	0.485	0.560	0.428	0.628	0.335	0.956	0.879	0.699	0.424

0.103	0.176	0.387	0.300	0.379	0.276	0.179	0.321	0.192	0.250
0.333	0.384	0.564	0.587	0.857	1.064	0.698	0.621	0.232	0.316
0.421	0.309	0.654	0.729	0.228	0.529	0.832	0.935	0.452	0.426
0.266	0.750	1.056	0.936	0.911	0.820	0.723	1.201	0.935	0.819
0.225	0.326	0.643	0.337	0.721	0.837	0.682	0.987	0.984	0.849
0.187	0.586	0.529	0.340	0.829	0.835	0.873	0.945	1.103	0.710
0.153	0.485	0.560	0.428	0.628	0.335	0.956	0.879	0.699	0.424

Tables

Precise comparison

- Good for **multivariate** variables
- Rows should invite **comparison** (e.g. sort table in ascending order by one column value)
- The order of columns represents their **importance**

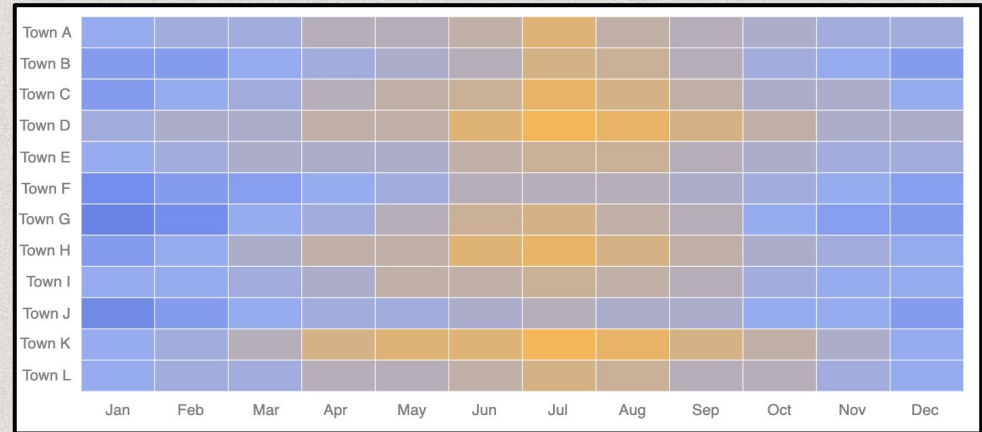
Frequency (Hertz)	Scientific Pitch	Helmholtz/ German	Octave Name	Pipe Length
8.176	C-1	CCCC		64'
16.352	C0	CCC	sub-contra	32'
32.703	C1	CC	contra	16'
65.406	C2	C	great	8'
130.81	C3	c	small	4'
261.63	C4	c'	1-line	2'
523.25	C5	c''	2-line	1'
1046.5	C6	c'''	3-line	1/2'
2093.0	C7	c''''	4-line	
4186.0	C8	c'''''	5-line	
8372.0	C9	c''''''		

Tips: change background color, font size, alignment of values, add icons (e.g. ) to emphasise content.

Tables

Heatmaps

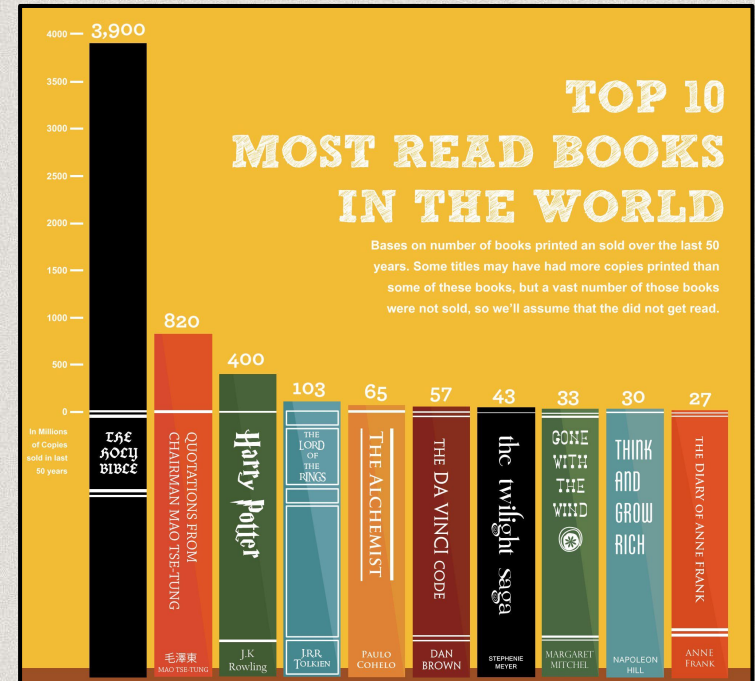
- Good for **multivariate** (3) variables
- Numeric values are mapped to colors
- It shows the **variance** (gradation) across values and can highlight patterns (e.g. correlation between month and temperature)
- Not good for precise comparison



Graphs

Bar graph

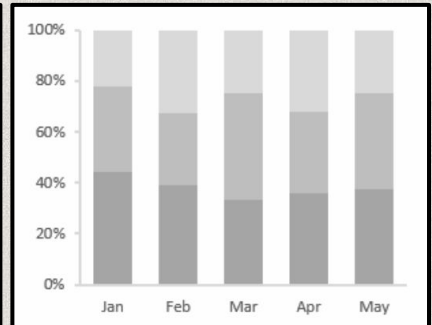
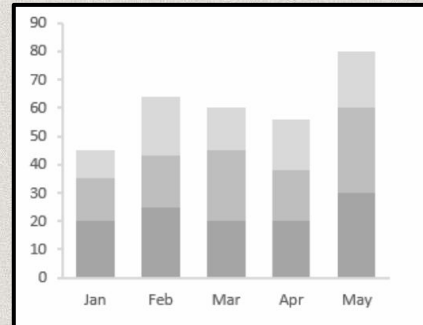
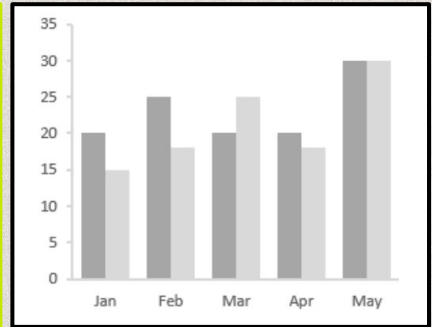
- Good for **bivariate** (2) analysis, often including
 - 1 categorical or ordinal variable
 - 1 numeric variable
- Good for **comparison** between (distribution of) categories



Graphs

Bar graph

In column graphs the categorical/ordinal variable is **independent** (x axis), while the numeric one is **dependent** (y axis)

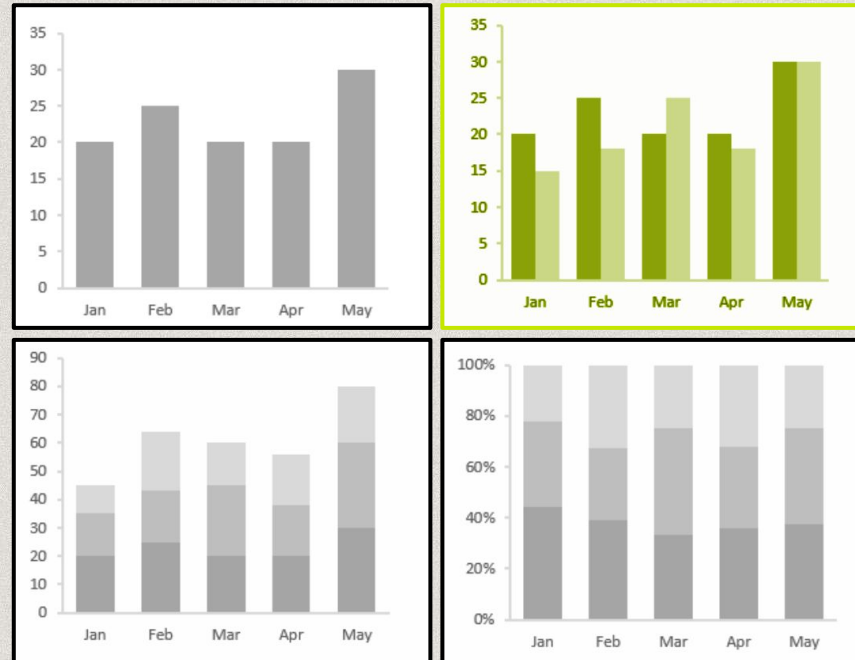


Graphs

Multi-series bar graph

When we need to compare sub-categories falling under a macro-category, we have **multi-series** or grouped bar graphs.

Sub-categories are differentiated by colors.

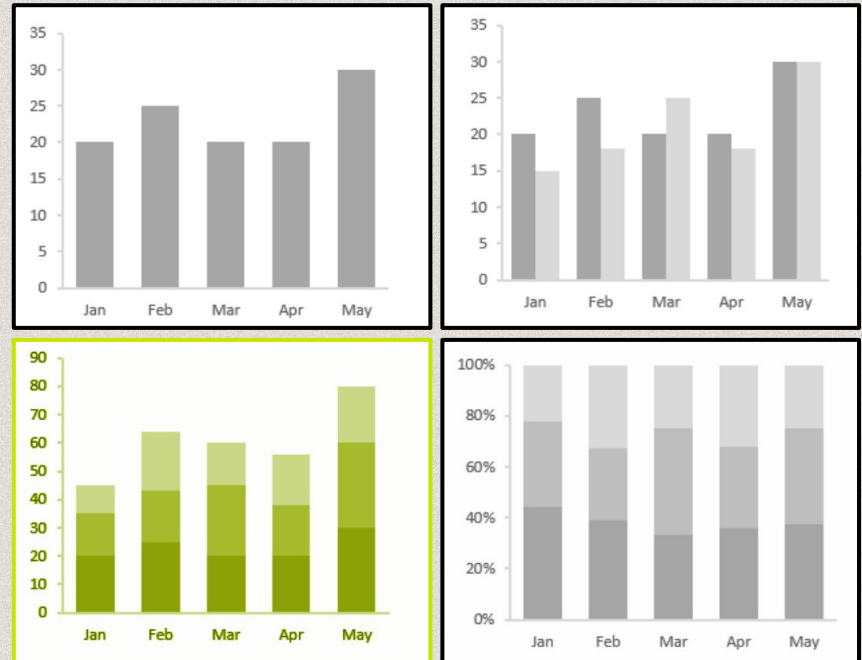


Graphs

Stacked bar graph

Similarly to grouped bar graphs, **stacked bar graphs** allow to compare sub-categories. These are placed on top of each other rather than next to each other.

Sub-categories are differentiated by colors.

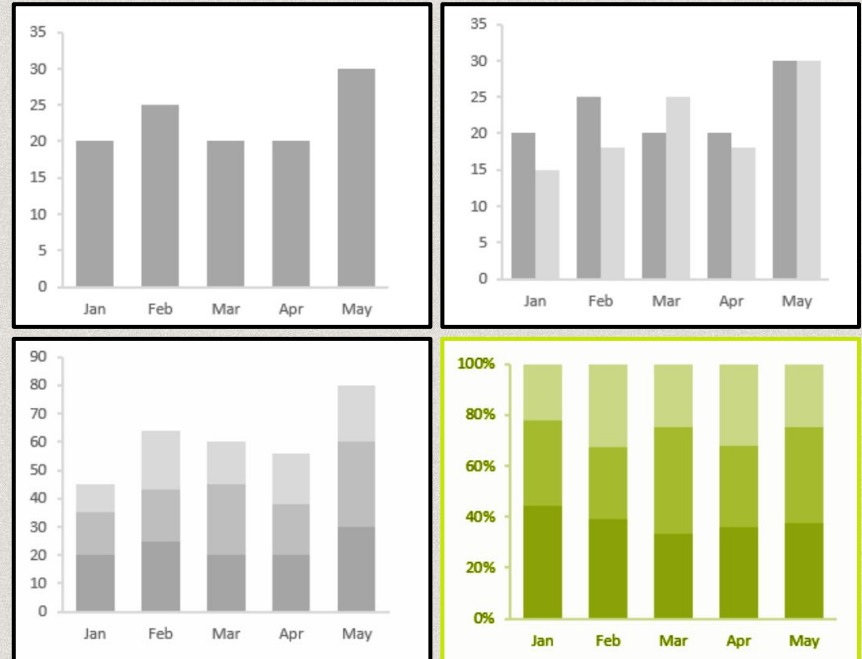


Graphs

100% Stacked bar graph

100% stacked bar graphs or normalised stacked bar graphs, allow to compare the contribution of sub-categories to the category they belong to (proportion).

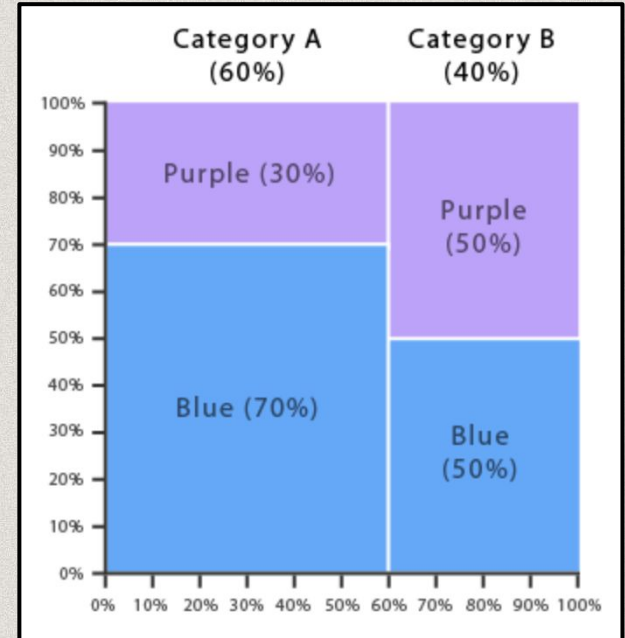
Sub-categories are differentiated by colors.



Graphs

Marimekko

A special 100% stacked bar graph, wherein not only the height but also the width of columns grows according to the proportion of sub-categories wrt the macro-category.

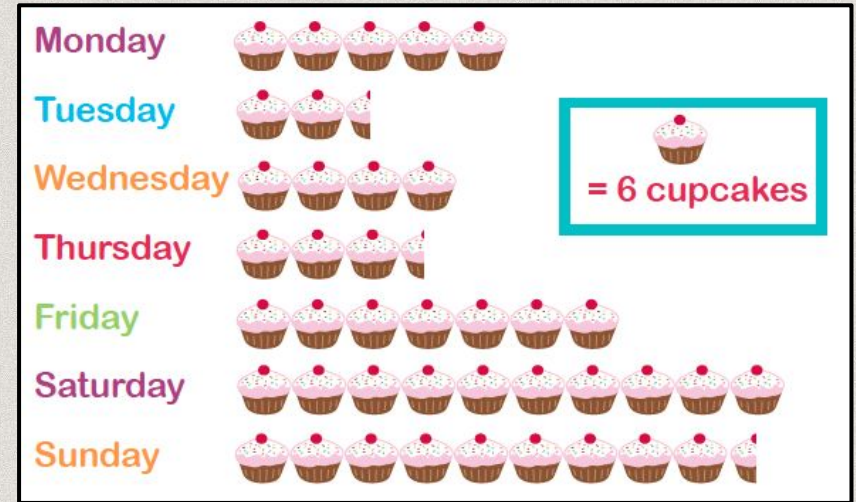


Graphs

Pictograms

Pictograms are special bar graphs where columns are replaced with **icons**.

It is a more engaging way to invite **comparison**.

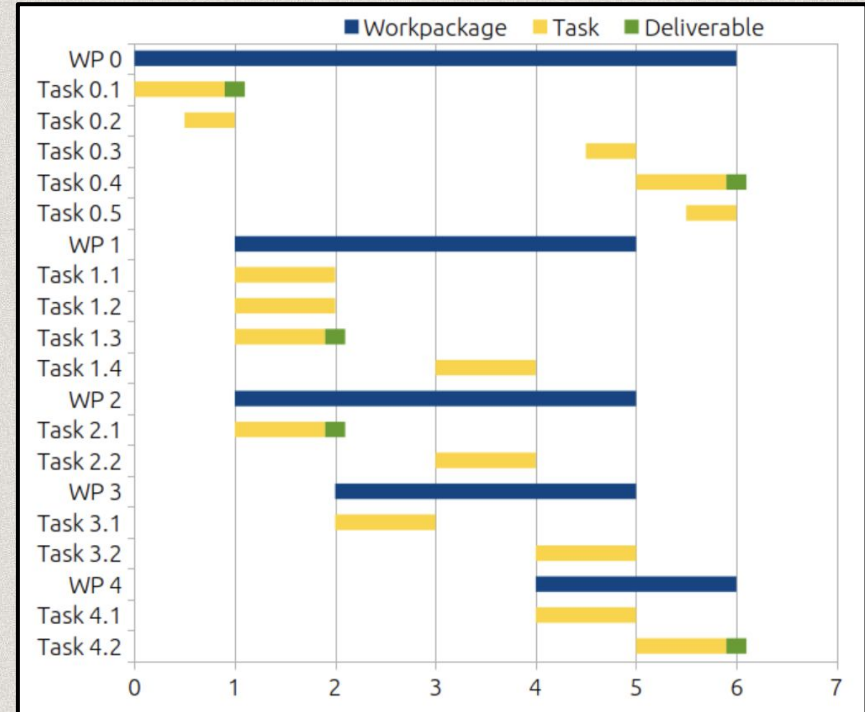


Graphs

Gantt chart

Halfway between a table and a bar chart (wherein the origin of axes may change for every column), represents variables over a time span (the ordinal value on the x axis).

It allows to see **dependencies** between variables.

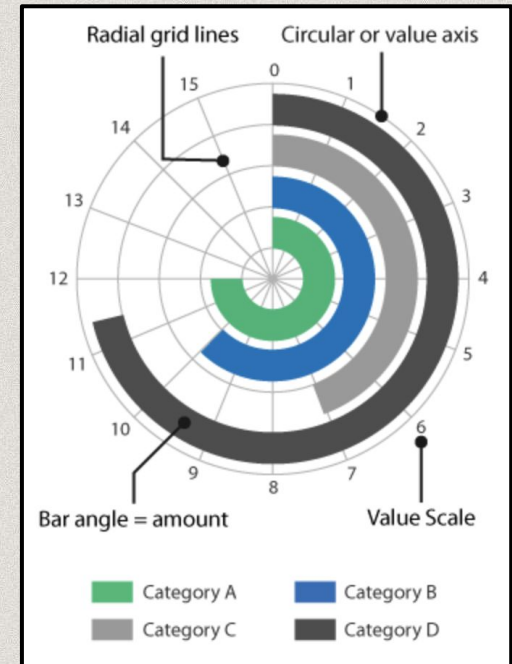


Graphs

Radial bar chart

A bar chart where the cartesian system is replaced by a **polar system**.

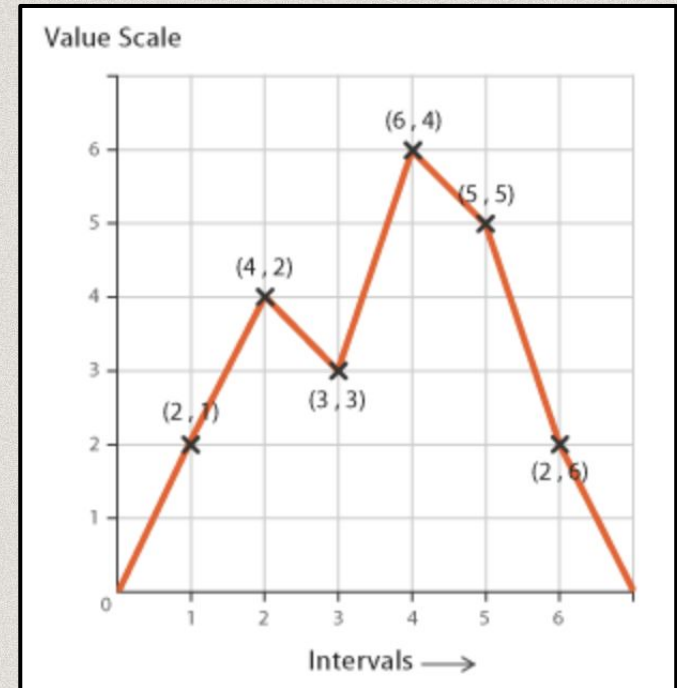
Unfortunately, they are easy to be **misread**. Outer bars generally tend to seem bigger than inner bars, while it does not imply that is true (see the example).



Graphs

Line graphs

- Good for **bivariate** analysis
- Like in bar graphs, one variable can be categorical/ordinal
- Variables are always **dependent**
- For each category only one point is shown. The line can be drawn or not
- Good to show **time-series** and highlight **trends**.



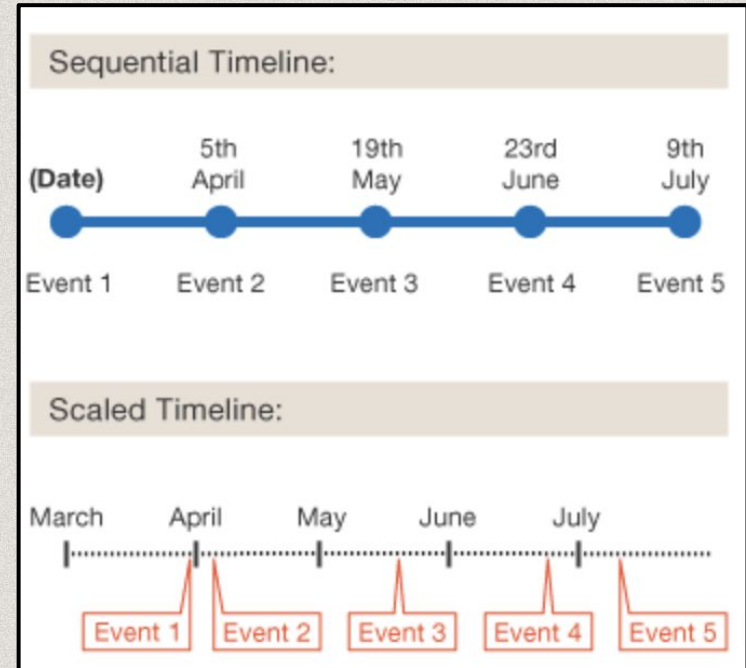
Graphs

Timeline

Data points can be displayed on a vertical/horizontal line.

The distance between points can be according to a **scale** or in **sequence** (no scale).

In a scale-based timeline it's possible to see the **clusters** of data points.



Data visualisation sense making

02

Test

Answer the questionnaire (20 minutes)

<https://forms.gle/kchdc6fbP9CT82wx7>

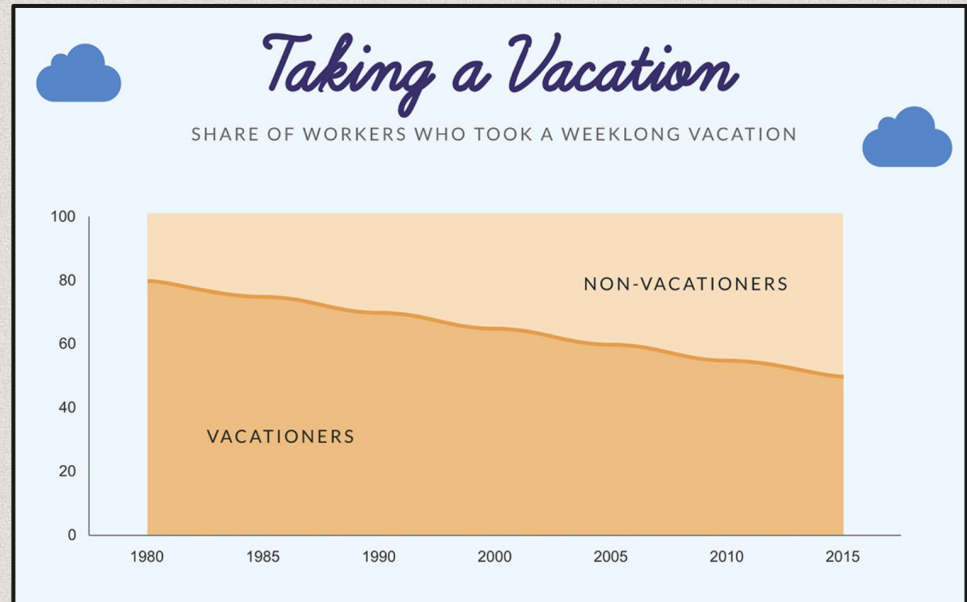
Graphs

Area charts

A special line chart where colors highlight the **dependency** between 2 variables.

It both shows the **trend** and the **size** of a phenomenon.

Values are scaled (e.g. percentage).



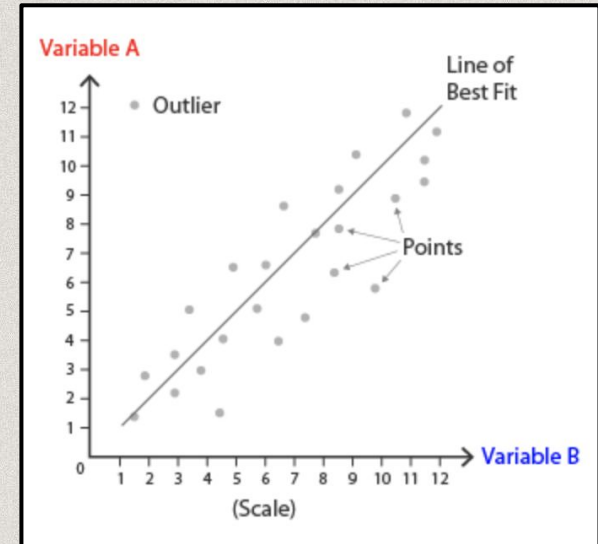
Graphs

Scatter plots

- Allow **bivariate** analysis between numeric values.
- Show the **correlation** between two numeric variables and the **outliers**.

Correlation is a measure that determines the degree to which the movement of two different variables is associated.

If a correlation exists, a line (called **best fit**) can be drawn.

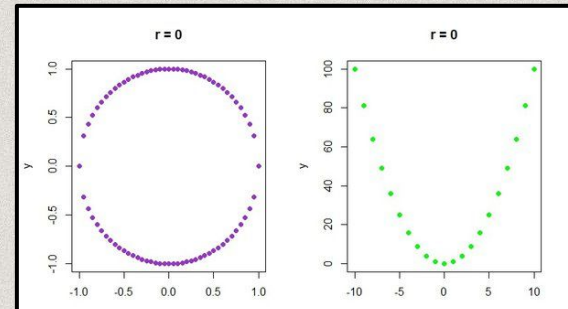
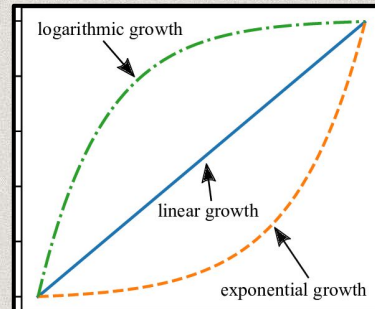
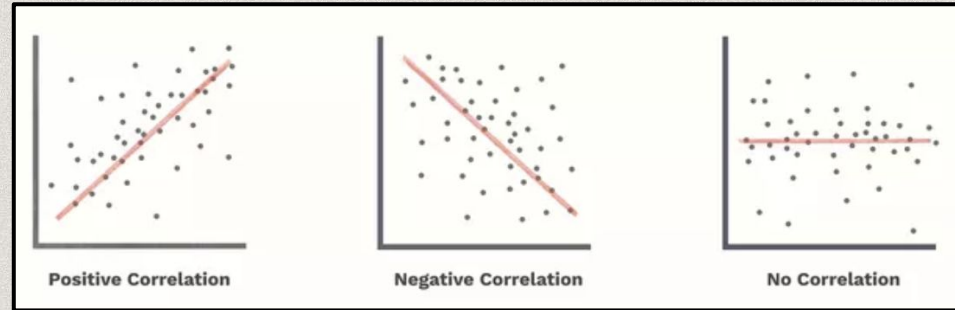


Graphs

Scatter plots

Patterns (highlighted by a **best fit** line) include:

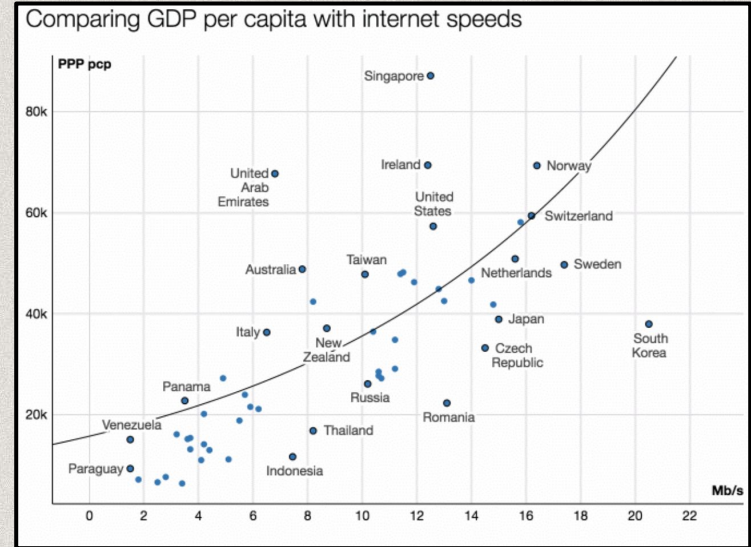
- positive (values increase together),
- negative (one value decreases as the other increases),
- null (no correlation),
- linear
- exponential
- U-shaped.



Example

Positive correlation

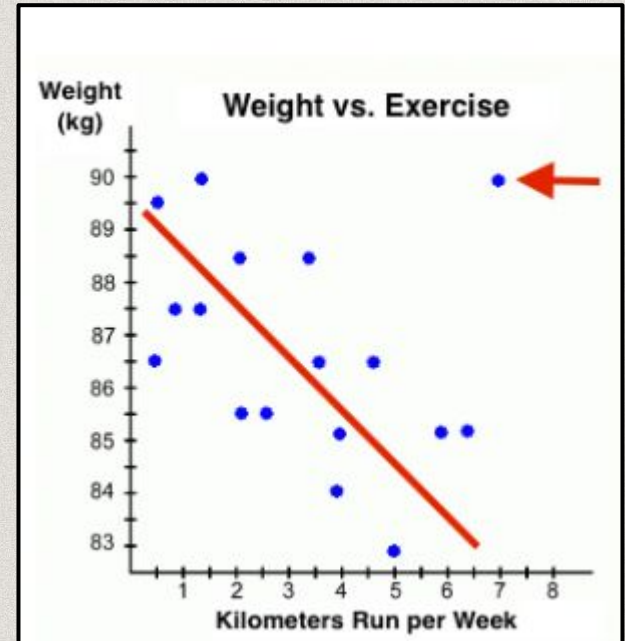
The more the income of a country grows (y axis), the more the speed of wifi grows (x axis) - and vice versa.



Example

Negative correlation

The more a person exercise (x) the less a person weights (y).

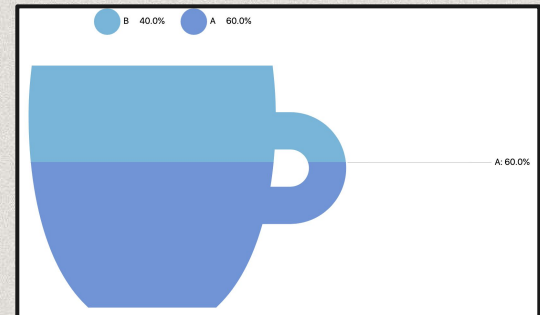
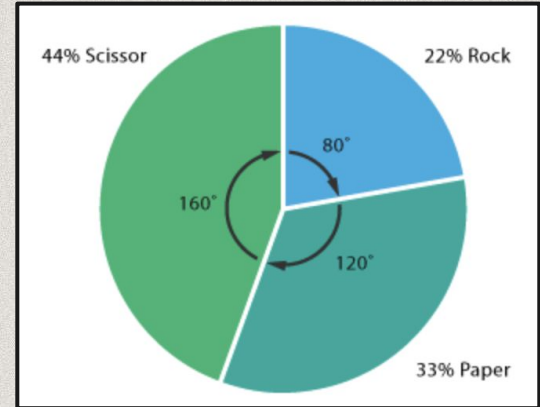


Charts

Pie chart

- Shows the **proportion** of parts (categorical values) of a whole.
- Good for **univariate** analysis with **dependent** values (if one changes also others change).
- **Angles** do not allow a precise comparison

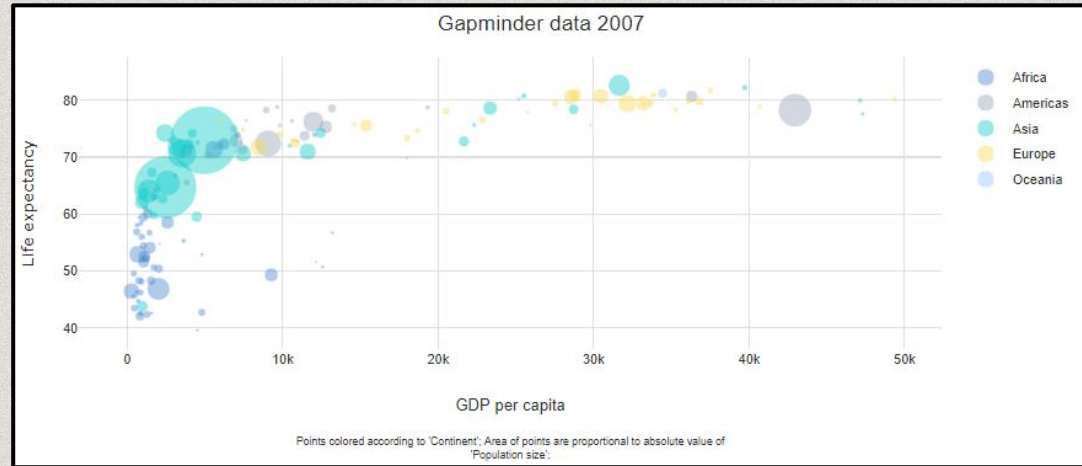
Circles can be replaced with images.



Charts

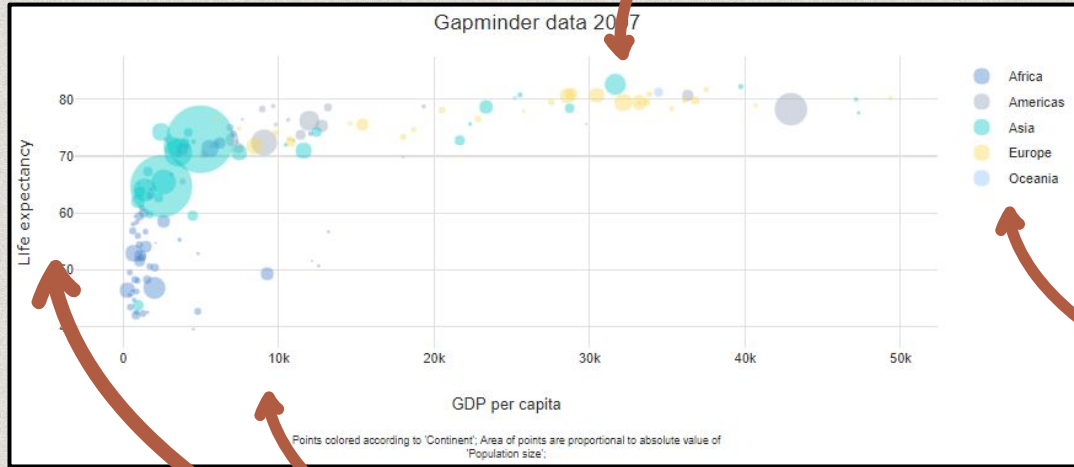
Bubble chart

- Good to compare a great amount of **multivariate** data
- Position and proportion can reveal **correlation** (e.g. here is logarithmic)



Charts

Bubble chart



The area is a numerical value

Points are labelled with a categorical value (country)

The color of points can represent another categorical value (continent)

X and Y are numerical values

Wordle

- the size of words represents a **univariate** quantitative aspect of categories.
- The layout is based on a randomised greedy algorithm that positions words so that they fill empty spaces and words do not overlap.

Tag clouds can be **misleading** (long words, letters with ascending lines seem bigger) and do not allow precise comparison.

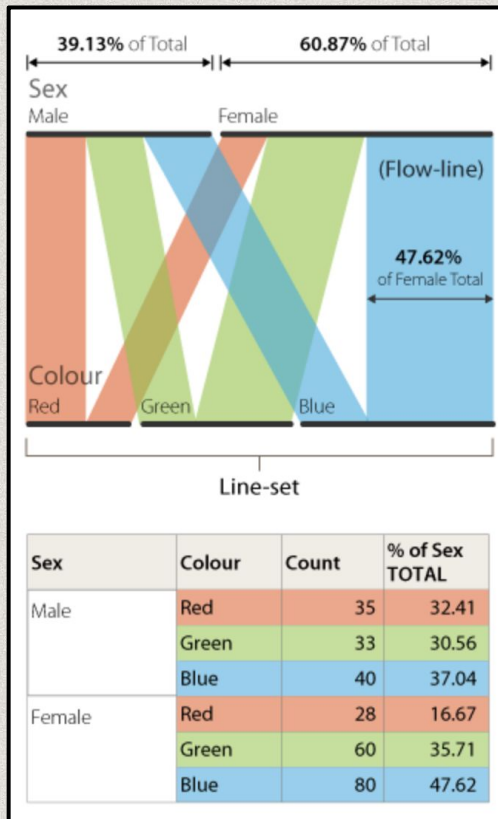


Charts

Parallel sets

Allow to compare **proportions** and relations between **multivariate** categorical values.

Relations are shown as flows between pairs of values.

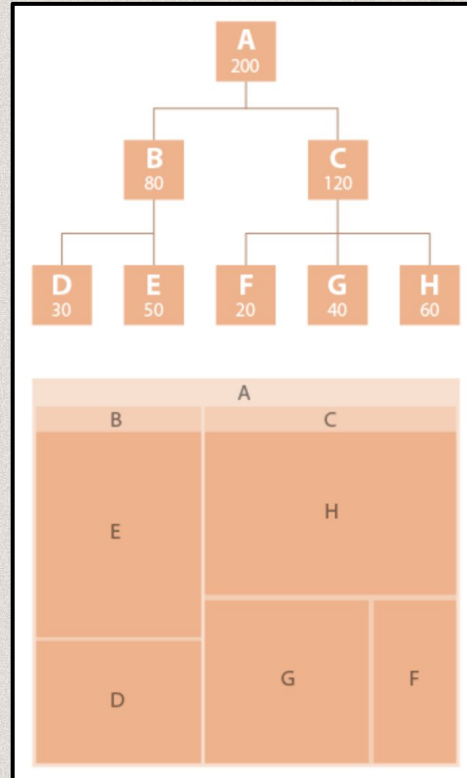


Maps

Tree map

Like pie charts, a tree map shows the **proportion** of parts of a whole.

It is a different way to display distribution in a **hierarchy**.

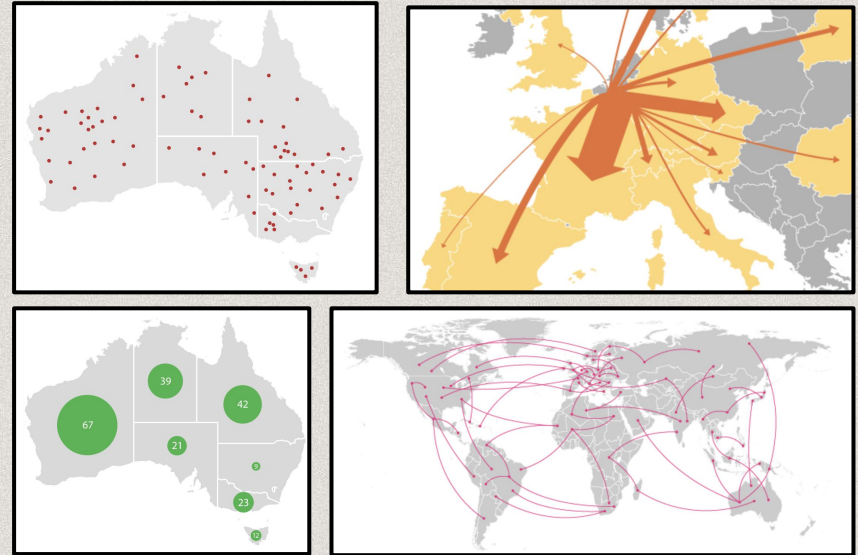


Maps

Geographical map

Data that can be **geo-localised** can be plotted on a map.

Maps can be combined with other visual strategies (e.g. flows, trajectories, dots, bubbles).



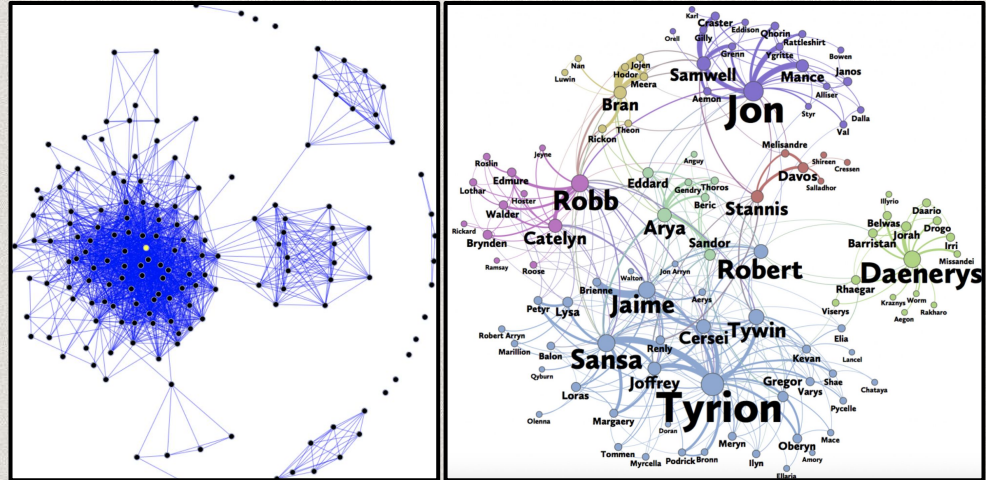
Networks

Graph networks

Categorical data are mapped to **nodes** and relations (co-occurrence) become (weighted) **arcs** between nodes.

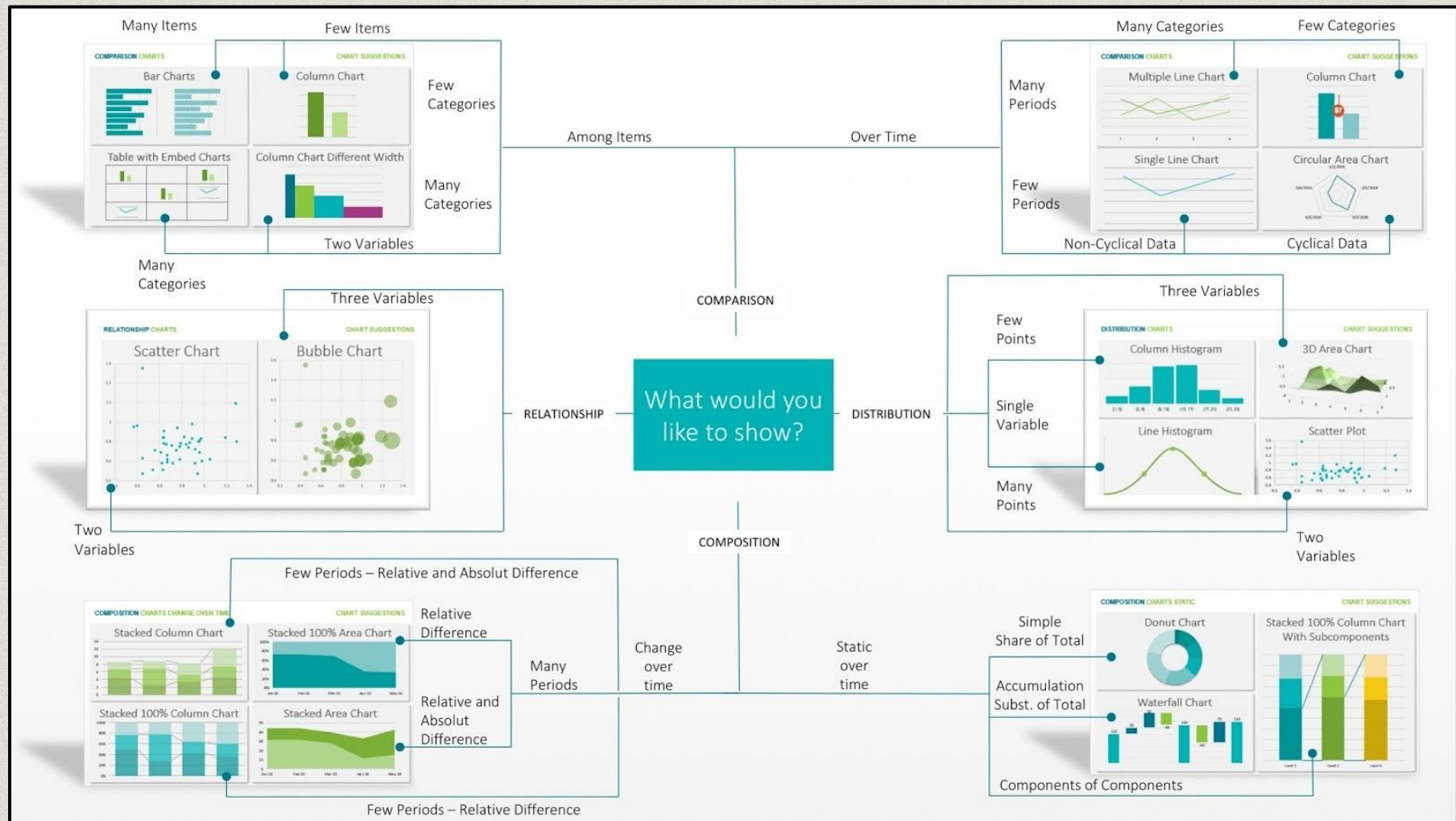
The location (closeness) of clusters is relevant, showing **similarities** between groups of data.

When too big, they are often unreadable.



02

How to decide?



Discuss

How to decide?

Some scenarios

1. The more classes a student misses, the more likely their grades are to decrease.
2. The number of times you look at your phone over the day.
3. The proportion of fantasy books on your shelf.

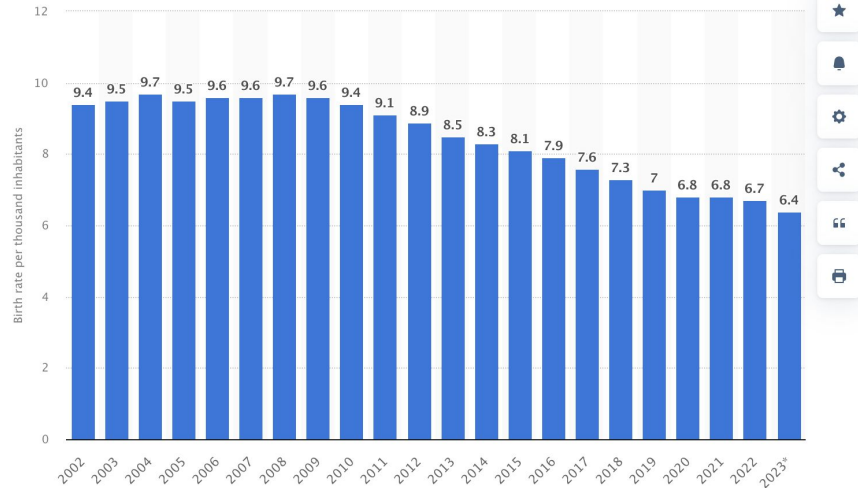
Which pattern would you highlight?
Which chart would you choose?

Discuss

Society › Demographics

Birth rate in Italy from 2002 to 2023

(per 1,000 inhabitants)



[Additional Information](#)

© Statista 2025

[Show source](#)

How to decide?

*Why do you think it happens?
What other features would you
explore to understand this
pattern?*

Discuss

How to decide?

Italian mothers older and older

Similar to citizens of other European countries, Italians also postpone parenthood to a later age. While the average age of an Italian mother at childbirth in the 1990s was 29.9 years, in 2022 females giving birth were roughly 32.4 years.

Italy, a country with one of the lowest fertility rates in the world

If compared with the fertility rates around the world, Italy was one of the 20 countries which registered the lowest fertility rate in 2023. The leader of the global ranking was Taiwan, where only 1.09 babies were born per woman.

Do two hints make an evidence?

TO DO

Next week

Come prepared

- Bring your laptop if possible
- Install Python3 and a IDE (e.g. PyCharm)
- Know how to run a py script via shell (read [here](#))
- Know how to install a Python library with pip (install [RDFlib](#))

Alternatively:

- Learn how to access and create a notebook on Colab, and how to install/import libraries there.



EDA

Thanks!

Do you have any questions?

marilena.daguino2@unibo.it

https://github.com/marilenadaquino/information_visualization

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